

Simon Says Memory Game: Course Project Summary

Tech 60: Introduction to Electronics | Team: SSMG | Jared Lomeli, [REDACTED]

Overview

Our project is a version of Simon Says built on the Arduino UNO R3 from the ELEGOO Starter Kit. The board flashes a pattern of four colored LEDs with a click from a buzzer, and the player needs to repeat it with a matching combination of the four push buttons. Each round adds one random color and plays slightly faster than the last. The game ends when the user presses the wrong button or there is no press received by the game after 3 seconds.

How It Works

Inputs. Each button connects a digital pin to ground. The Arduino's internal pull-up resistor holds the pin HIGH, so a press pulls it LOW with no external resistor needed. The code rechecks a press after 20 ms (debouncing) and waits for release so one press is never counted twice.

Processing. The ATmega328P stores the pattern in an array of up to 50 steps. On each round, it replays the whole array of previous colors, then adds an additional random color to the end of the sequence. Finally, it compares every player press against the whole array of stored previous colors in order.

Outputs. In the first round, each light stays on for half a second (500 ms). Every round after that, each light flashes 25 ms quicker, until they reach their fastest speed of 200 ms at round 13. In addition, each round adds one more color to the color sequence that the user must match.

Part	Pin	What it does
4 LEDs (with resistors)	D2-D5	Light up to show the pattern
4 push buttons	D6-D9	Player input
Buzzer	D10	Beeps on each color at game over
A0	A0	Makes each game's pattern random

Program Flowchart



